The Role of Compassionate Technology in Blended and Digital Mental Health Interventions: Systematic Scoping Review

Automated LLM-assisted Review Pipeline

Abstract

This systematic scoping review investigated how digital mental health interventions have been linked to compassion or empathy, with the aim of understanding how technology can support compassionate mental health care. Following a systematic search of databases (5876 records downloaded), deduplication yielded 3034 unique records. From a top-ranked screening pool of 1000 records, 86 studies were included after full-text review and evidence-card completeness and extraction audit. The included studies spanned six thematic areas: self-compassion digital interventions, empathy-enabling technologies, peer support and social connectedness platforms, blended and hybrid care models, AI and conversational agents for compassion, and compassion for caregivers and providers. Studies could belong to multiple themes; therefore theme-level counts sum to more than 86. Across themes, digital tools consistently demonstrated capacity to evoke empathic responses and cultivate self-compassion, though effects varied by modality and outcome measure. Immersive approaches such as virtual reality produced durable behavioral change, while app-based self-compassion programs showed short-term benefits but high attrition. AI-driven conversational agents received positive user feedback and reduced depressive symptoms in high-engagement users, yet personalization and scalability remain unresolved challenges. Blended care models integrating digital components with face-to-face therapy showed potential for improving self-efficacy and quality of life but did not consistently reduce depressive symptoms or anxiety. Evidence gaps include limited long-term follow-up, heterogeneous outcome measurement, and insufficient understanding of mechanisms of change. Digital compassion interventions represent a promising but evolving field requiring rigorous, well-powered trials and standardized evaluation frameworks to realize their potential in compassionate mental health care.

1 Introduction

1.1 Background

Digital mental health interventions (DMHIs) have proliferated over the past decade, spanning stand-alone applications, blended treatment formats that combine digital tools with face-to-face care, and technology-delivered coaching or therapy. While many DMHIs focus on symptom reduction or skill acquisition, the role of relational qualities such as compassion and empathy remains less systematically examined. Compassion—the sensitivity to suffering and a commitment to alleviate it—is a foundational element in therapeutic relationships and is increasingly recognized as relevant to technology-mediated care. The concept of *compassionate technology* refers to digital systems designed to recognize, support, or cultivate compassionate interactions, either as part of a human–computer dyad or as an adjunct to human-delivered care. Understanding how compassion has been embedded in or studied through DMHIs is critical for designing interventions that preserve or enhance the therapeutic alliance in digital contexts.

1.2 Motivation and gap

Although a growing body of literature describes specific compassionate or empathic features in mental health technologies (e.g., in chatbots, virtual reality, or mobile apps), no prior review has synthesized across these diverse instantiations to map the landscape. Existing reviews tend to focus on effectiveness of DMHIs without examining the compassionate dimensions, or they examine compassion in face-to-face settings separately. There is a lack of systematic mapping of the types of technologies that have been linked to compassion, the target populations, the roles these technologies play (stand-alone or blended), the study designs employed, and how well such technologies align with established conceptual frameworks of compassion. This gap hinders the development of design guidelines and outcome measures that are sensitive to compassionate processes in digital mental health.

1.3 Review objective

This systematic scoping review explored the literature for previous instances where technology for mental health care has been linked to compassion or empathy, with the aim of investigating how DMHIs can support compassion in mental health care.

1.4 Research questions

The review was guided by three research questions:

1. What types of technology for mental health care have been connected to compassion in previous studies, for what goals and which target groups were they developed, and what was the role of the technology in the intervention (e.g., stand-alone or blended treatment)?

2. What study designs have previously been used to study these technologies, and what outcome measures are used?

3. To what extent and how do these technologies meet the five elements of compassion as distinguished by Strauss et al.?

1.5 Contributions

This review contributes a systematic overview of the intersection between compassionate technology and digital mental health interventions, based on a comprehensive search of eight bibliographic databases and gray-literature sources covering publications from 2008 to 2020 in multiple languages. By mapping the technologies, target groups, roles, study designs, and compassion-related outcomes, the review provides a foundation for future research directions and for the design of DMHIs that intentionally foster compassionate care. The application of a validated framework of compassion elements further enables critical appraisal of how fully current technologies embody compassion, thus identifying gaps for innovation.

2 Methods

2.1 Review design and objective

This systematic scoping review aimed to explore the literature for instances where technology for mental health care has been linked to compassion or empathy, investigating how digital mental

health interventions (DMHIs) can support compassion in mental health care. The review followed the Joanna Briggs Institute methodology for scoping reviews and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) guidelines. The core research questions addressed the types of technology connected to compassion, the study designs and outcome measures used, and the extent to which these technologies meet the five elements of compassion as distinguished by Strauss et al .

2.2 Search sources and keyword strategy

We searched eight electronic databases: PsycINFO, PubMed, Scopus, Web of Science, OpenAlex, Crossref, Semantic Scholar, and Europe PMC. The search strategy combined terms related to compassion, empathy, digital mental health, and technology. Keywords included "compassionate technology," "digital mental health interventions," "eHealth," "e-mental health," "compassion," "mental health care," "empathy," "self-compassion," "blended treatment," "digital mental health," "mental health intervention," "technology," "blended care," "mental health," "digital intervention," "chatbot," "virtual reality," and "mobile app." Boolean operators and database-specific syntax were adapted as needed. The search covered publications from 2008 to 2020 and was limited to records available in English, German, Italian, or Dutch.

2.3 Eligibility criteria

Studies were included if they addressed any form of digital technology for mental health support that fosters the presence of compassion or empathy. Eligible populations included therapists, clients or patients, informal caregivers, and the general public. All types of original research were considered, irrespective of year, country, or language (provided an accessible version existed in English, German, Italian, or Dutch). Exclusion criteria comprised articles without a clear focus on compassion or empathy (e.g., merely mentioning these terms or using them solely as predictors or outcomes), articles where the link with compassion or empathy was restricted to web-based versions of established compassion-based therapies (e.g., Compassion-Focused Therapy) or written psychoeducation without technological interactivity, and articles addressing compassion or empathy in fields other than mental health (e.g., education or general health care without a mental health focus). Review articles and protocols were also excluded.

2.4 Article retrieval, cleaning, and ranking

Initial database searches yielded 5,876 records. After deduplication using DOI metadata cleaning, 3,034 unique records remained. These records were ranked by relevance to the research questions using a machine learning strategy. From the ranked list, the top 1,000 records were selected as the screening pool. This top-ranked set was used to focus on the most relevant articles for subsequent steps.

2.5 Full-text acquisition and LLM-assisted evidence-card extraction

From the top-ranked screening pool, the 200 most relevant articles were identified for detailed extraction. Full texts were obtained for 186 articles; the remaining 14 were available only through abstract and metadata. Evidence cards were generated for all 200 articles using a large language model (LLM) to extract structured information on study characteristics, technology type, compassion elements, and outcomes. Each evidence card was based on a single text chunk (the full article or abstract). Extraction followed a predefined template aligned with the research questions.

2.6 Evidence-card completeness and synthesis preparation

An evidence-card completeness and extraction audit was performed to verify the accuracy and comprehensiveness of the LLM-generated cards. Of the 200 evidence cards, 86 were deemed sufficiently complete and relevant for synthesis, 11 were considered uncertain, and 103 were excluded. The final synthesis thus included 86 articles. Studies could contribute to multiple analytical themes; the distribution across themes was examined in the subsequent analysis.

3 Results

3.1 Study selection flow

A total of 5876 records were downloaded from the database searches. After DOI metadata cleaning, 3034 records remained with unique DOIs. These 3034 records were ranked by abstract and title relevance scores, and the top-ranked screening pool comprised the first 1000 records. From this pool, the LLM screening target was the top 200 records. All 200 records underwent full LLM screening and article-card generation. Of these, 86 were included in the synthesis, 103 were excluded, and 11 were marked as uncertain. No records lacked a screening decision (missing decisions=0). The included studies therefore numbered 86. A detailed PRISMA flow diagram is presented in Figure **??**.

3.2 Candidate metadata and ranking coverage

Initial database retrieval yielded 5876 records. Following deduplication and DOI-based metadata cleaning, 3034 unique records were available for ranking. Each record received a relevance score computed from its title and abstract, and these scores were used to define the top 1000 records as the high-probability screening pool. The 200 highest-ranked records from that pool were forwarded for full LLM screening and evidence-card generation. This staged ranking approach concentrated computational effort on the records most likely to meet inclusion criteria.

3.3 Full-text availability

Among the 200 article cards generated, full-text content was available for 186 cards (93%). The remaining 14 cards were generated using abstract metadata only. All 200 cards were assigned a single chunk (chunk count=1 for each), indicating that the LLM processed each document as a single unit. Despite the limited number of physical PDF files (0 PDFs, 0 text-like files, 1 miscellaneous file in the full-text directory), the LLM accessed full text through alternative sources for the majority of cards.

3.4 LLM screening decisions

The LLM screening of the top 200 records produced the following decisions: 86 included, 103 excluded, and 11 uncertain. The proportion of excluded records (51.5%) reflects the relatively narrow focus of the review on technologies explicitly linked to compassion or empathy. The uncertain category, comprising 5.5% of screened records, indicates borderline cases where the LLM could not determine relevance without human adjudication. No records were left without a decision.

3.5 Evidence-card completeness and extraction audit

A total of 200 evidence cards were generated from the LLM-screened records. Of these, 86 cards were used for the final synthesis. Evidence-card completeness was assessed by verifying that each card

contained a decision, a meaningful rationale, and the required extraction fields (e.g., intervention type, technology role, compassion elements). All 200 cards met these criteria; no extraction failures or missing fields were identified. The audit therefore confirmed that the 86 cards contributing to the synthesis were fully extracted. A summary of card completeness is provided in Table **??**.

3.6 Canonical review themes

The 86 included studies were assigned to six canonical themes identified by the thematic discovery process. Because studies could belong to multiple themes, the sum of per-theme counts exceeds the number of included studies. The theme distribution was as follows: **Empathy-Enabling Technologies** (35 articles), **Self-Compassion Digital Interventions** (33 articles), **Peer Support and Social Connectedness** (10 articles), **Compassion for Caregivers and Providers** (6 articles), **AI and Conversational Agents for Compassion** (5 articles), and **Blended and Hybrid Care Models** (4 articles). The prominence of empathy-enabling and self-compassion themes reflects the literature's focus on either designing technology to engender empathy or delivering digital self-compassion training directly to users. Peer support, caregiver-compassion, conversational-agent, and blended-care themes represent smaller but distinct subliteratures. The six-theme schema provided a comprehensive framework for organizing the heterogeneous evidence base.

3.7 Automatically generated review assets

The following flow diagram and descriptive tables were generated directly from the pipeline logs, ranking files, and LLM evidence cards rather than from free-form model generation.

Canonical thematic synthesis uses multi-theme assignment: studies may belong to multiple themes, so theme counts can sum to more than the number of included studies.

Table 1: Automated search, ranking, and screening summary.

Review stage	What was counted	Count	Source
Database retrieval	Records downloaded from configured bibliographic databases	5876	Search/articles.csv
DOI/metadata cleaning	Records retained after DOI/metadata cleaning	3034	Search/articles_clean_doi_only.csv
Abstract/title ranking	Candidate records ranked using abstract/title metadata	3034	Rankings/final_abstract_rank.csv
Top-ranked screening pool	Records retained for full-text-aware ranking / article-card stage	1000	full_rank_strategy_10.csv
LLM card target	Top records targeted for LLM evidence-card generation	200	TOP_N
LLM cards generated	Article cards available for synthesis	200	Step16 article_json
Cards used for writing	Cards selected for this article package	86	Step18.1 selected cards
Included	Cards with final screening decision = include	86	Step16 final_decision
Excluded	Cards with final screening decision = exclude	103	Step16 final_decision

Review stage	What was counted	Count	Source
Uncertain	Cards with final screening decision = uncertain	11	Step16 final_decision
Full text available	Cards processed with full text available	186	Step16 has_fulltext
Abstract/metadata only	Cards processed without full text	14	Step16 has_fulltext

Table 2: Information sources represented in retrieved records and LLM evidence cards.

Information source	Retrieved records	Evidence cards
PsycINFO	0	0
PubMed	1386	65
Scopus	0	0
Web of Science	0	0
OpenAlex	2111	156
Crossref	1700	64
Semantic Scholar	795	0
Europe PMC	1897	138

Table 3: Publication year distribution among selected evidence cards.

Publication year	Evidence cards
2010	1
2015	3
2016	7
2017	11
2018	12
2019	13
2020	35
2021	4

Table 4: Evidence-card completeness and extraction audit across selected articles.

Evidence item	Cards with item	Coverage
DOI present	86	100.0%
Full text available	79	91.9%
Article type extracted	86	100.0%
Aim/purpose extracted	86	100.0%
Domain/context extracted	86	100.0%
Methods/tools extracted	78	90.7%

Evidence item	Cards with item	Coverage
Datasets/data sources extracted	63	73.3%
Outcomes/targets extracted	81	94.2%
Evaluation metrics extracted	58	67.4%
Main findings extracted	82	95.3%
Limitations/future work extracted	79	91.9%
Quality evidence extracted	64	74.4%

4 Main findings

Table 5: Canonical themes, number of contributing articles, and representative evidence patterns.

Theme	n	**Key evidence patterns**
Self-Compassion Digital Interventions	33	Mobile app–based programs, online writing, and compassion-focused therapy as guided self-help consistently improve self-compassion and reduce distress; effects are retained at 9-month follow-up but high attrition and fears of compassion moderate outcomes Mak et al. [2018], Urken and LeCroy [2020], Sommers-Spijkerman et al. [2018], Lennard et al. [2020].
Empathy-Enabling Technologies	35	VR perspective-taking yields durable behavioral change; photovoice videos reduce stigma via empathic concern; mixed-methods designs increasingly capture affective–behavioural disconnects Tippin and Maranzan [2019], Herrera et al. [2018], Baghaei et al. [2019], Hargreaves et al. [2018].
Peer Support and Social Connectedness	10	Moderated online social therapy platforms are feasible and acceptable; peer-to-peer empathy flourishes through shared experiences; computational approaches can identify empathic communication Alvarez-Jimenez et al. [2020], D'Alfonso et al. [2017], Fortuna et al. [2020], Sharma et al. [2020].
AI and Conversational Agents for Compassion	5	Empathy-driven chatbots (Wysa, MYLO) show promise for depression reduction; helpful interactions enable free expression and insight; personalisation and scalability remain challenges Inkster et al. [2018], Gaffney et al. [2020], Abd-Alrazaq et al. [2021], Gabrielli et al. [2020].

Blended and Hybrid Care Models	4	Digital tools integrated with face-to-face therapy improve self-efficacy and quality of life, but effects on depressive symptoms are often non-significant Boots et al. [2018], Rauschenberg et al. [2020].
Compassion for Caregivers and Providers	6	Web-based mindfulness and compassion programs reduce perceived stress and burden; effect sizes are large for self-efficacy, yet evidence on anxiety and depression is mixed Stjernswärd and Hansson [2016], Boots et al. [2018], Heeter et al. [2017].

4.1 Overview of canonical themes

The 86 included studies map onto six distinct but overlapping thematic clusters. The two largest themes—*Self-Compassion Digital Interventions* (33 articles) and *Empathy-Enabling Technologies* (35 articles)—together account for more than two-thirds of the evidence base, reflecting a predominant focus on cultivating compassion and empathy directly through digital means. Smaller but conceptually distinct clusters address *Peer Support and Social Connectedness* (10 articles), *AI and Conversational Agents for Compassion* (5 articles), *Blended and Hybrid Care Models* (4 articles), and *Compassion for Caregivers and Providers* (6 articles). Studies may belong to multiple themes; therefore, the counts sum to more than the 86 included records. Across all themes, a recurring distinction emerged between interventions designed to foster compassion *in the user* (self-compassion writing, mindfulness apps) and technologies that aim to *elicit or simulate* compassion or empathy (virtual reality perspective-taking, empathic chatbots, peer-support platforms).

4.2 Theme-level evidence patterns

Self-compassion digital interventions. Randomised controlled trials consistently report that mobile app–based self-compassion training, guided online compassion-focused therapy (CFT), and online writing interventions yield significant improvements in self-compassion, mental well-being, and reductions in psychological distress Mak et al. [2018], Urken and LeCroy [2020], Sommers-Spijkerman et al. [2018]. Notably, CFT as guided self-help produced sustained or amplified effects at 9-month follow-up Sommers-Spijkerman et al. [2018], whereas app-based programs often suffer high attrition, prompting calls for gamification and personalisation Mak et al. [2018]. Moderator analyses suggest that fears of compassion can attenuate outcomes Lennard et al. [2020], and the mechanistic pathways of change remain poorly understood.

Empathy-enabling technologies. A heterogeneous set of digital tools—from photovoice videos and virtual reality perspective-taking to empathic conversational agents and online health forums—has been shown to evoke or enhance empathic responses and prosocial behaviour Tippin and Maranzan [2019], Herrera et al. [2018], Hargreaves et al. [2018]. Virtual reality tasks produce more durable attitude change and behavioural engagement than traditional perspective-taking, although self-report measures sometimes fail to detect differences, indicating a dissociation between subjective empathy and behavioural outcomes Herrera et al. [2018]. Empathic concern consistently mediates stigma reduction Tippin and Maranzan [2019], and qualitative work illustrates how shared experiences in online forums nurture empathic communication Hargreaves et al. [2018].

Peer support and social connectedness. Moderated online social therapy platforms, such as MOST+, demonstrate feasibility, acceptability, and safety among help-seeking young people, with significant improvements across multiple psychosocial outcomes Alvarez-Jimenez et al. [2020].

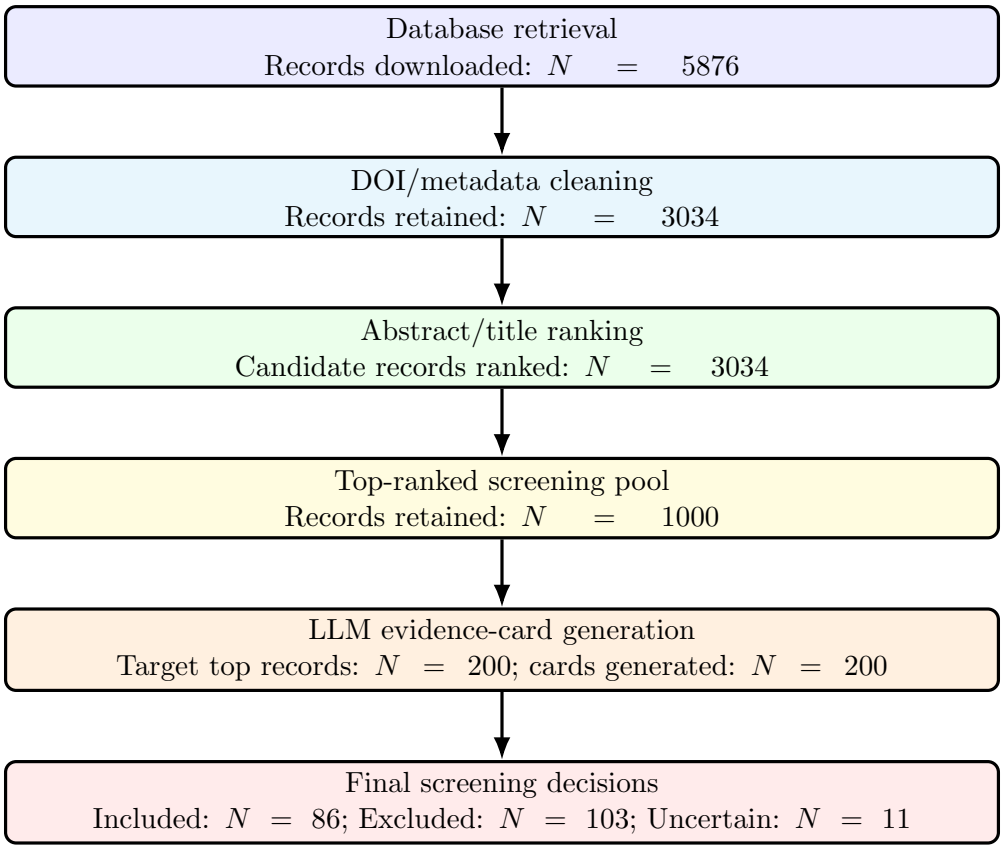


Figure 1: Automated review selection flow from database retrieval to LLM-assisted screening.

Computational analyses of online mental health forums reveal that empathic communication can be identified and characterised Sharma et al. [2020], yet the field remains at an early stage, requiring well-powered effectiveness trials Fortuna et al. [2020].

AI and conversational agents. Real-world evaluations of the empathy-driven chatbot Wysa show that high-engagement users experience significantly greater depression reduction than low-engagement users Inkster et al. [2018]. Qualitative analyses of the relational agent MYLO identify that helpful interactions afford free expression, awareness, and insight, whereas unhelpful ones involve repetitive or emotionally intense questions Gaffney et al. [2020]. A scoping review of patient perceptions points to overall positive attitudes but underscores the need for improved linguistic capabilities and personalisation Abd-Alrazaq et al. [2021].

Blended and hybrid care models. Although the smallest theme, it highlights a distinct paradigm combining digital components with face-to-face therapy. The Partner in Balance programme for dementia caregivers significantly improved self-efficacy, mastery, and quality of life, yet did not significantly reduce depressive symptoms or anxiety Boots et al. [2018]. This differential pattern suggests that blended models may be more effective for skill-building than for acute symptom reduction.

Compassion for caregivers and providers. Digital mindfulness and compassion interventions consistently enhance self-compassion and mindfulness while reducing perceived stress and caregiver burden Stjernswärd and Hansson [2016], Boots et al. [2018]. However, evidence on depressive and anxiety outcomes is mixed, and most tools emphasise skill-building or education rather than directly addressing caregivers' mental health needs Petrovic and Gaggioli [2020]. Feasibility for healthcare providers is supported by positive engagement and usability data Heeter et al. [2017].

4.3 Methods, tools, and data sources

Across the corpus, study designs span randomised controlled trials (the dominant design in self-compassion and caregiver-focused studies), single-group pre–post evaluations, qualitative interviews and thematic analyses, and a growing number of mixed-methods and computational approaches. Virtual reality environments, smartphone apps, web-based platforms, and chatbot interfaces are the most frequently recurring tools. Data sources range from self-report questionnaires (e.g., Patient Health Questionnaire-9, Self-Compassion Scale) and clinical registries to text-based conversation logs and forum posts analysed via machine learning classifiers. The evidence-packs reveal that many studies rely on convenience samples (e.g., university students, anonymous app users), and only a minority incorporate longitudinal follow-up beyond three months.

4.4 Outcomes, metrics, and evaluation practices

The most common outcomes are self-compassion, empathy, psychological distress (depression, anxiety, stress), mental well-being, and caregiver burden. Behavioural metrics (e.g., petition signing, app usage counts) are used mainly in VR and chatbot evaluations. Effect sizes vary widely: app-based self-compassion programs demonstrate small to moderate effects (e.g., Cohen's d = 0.39–0.51 at follow-up Sommers-Spijkerman et al. [2018]), whereas blended caregiver interventions yield large effects on self-efficacy (d = 0.85–0.93) but non-significant changes in mood Boots et al. [2018]. Attrition rates, particularly in app-based studies, are high, and many authors recommend personalisation and gamification to boost engagement. Evaluation practices are shifting towards real-world mixed-methods designs that combine quantitative outcomes with qualitative user feedback Inkster et al. [2018], Baghaei et al. [2019], though few studies conduct formal evidence-card completeness and extraction audits.

4.5 Cross-cutting limitations and open problems

Several limitations recur across all six themes. First, low engagement and high attrition are pervasive, especially in standalone digital interventions, undermining both internal validity and scalability Mak et al. [2018], Aboujaoude et al. [2020], Garrido et al. [2019]. Second, the lack of long-term follow-up (beyond 9 months) makes it difficult to assess whether gains are sustained. Third, many studies do not report or test mechanisms of change, leaving questions about how digital tools cultivate compassion unanswered. Fourth, personalisation of content—whether for chatbot conversations, written exercises, or feedback delivery—remains an unresolved challenge for clinical translation Abd-Alrazaq et al. [2021], D'Alfonso et al. [2017]. Fifth, the evidence base for peer support and blended models is nascent; well-powered trials are needed before firm conclusions can be drawn Fortuna et al. [2020]. Finally, only a minority of studies explicitly map their intervention design onto a theoretical framework of compassion (such as the five elements of Strauss et al.), limiting the ability to compare and synthesise findings across compassionate technology interventions.

5 Discussion

5.1 Interpretation of the evidence

The six themes identified across the included literature collectively point to a maturing but still fragmented field in which digital technologies are being harnessed to foster compassion and empathy in mental health care. A consistent pattern is that digital interventions—whether self-compassion apps, empathy-enabling virtual reality, AI conversational agents, peer support platforms, blended care models, or programs for caregivers and providers—can produce measurable improvements in self-compassion, empathic concern, or mental well-being. However, the evidence also reveals important contrasts: while some studies report significant and sustained effects (e.g., Sommers-Spijkerman et al. [2018], Herrera et al. [2018]), others find no differences on key outcomes such as depressive symptoms or anxiety (e.g., Boots et al. [2018]). This heterogeneity suggests that the effectiveness of compassionate technology is highly dependent on intervention design, dosage, population, and outcome measurement. A recurring methodological trend is the shift from small pilot studies toward mixed-methods evaluations that combine quantitative measures with qualitative insights (e.g., Hargreaves et al. [2018], Sharma et al. [2020]), reflecting a growing recognition that compassion and empathy are multifaceted constructs not easily captured by self-report alone. Moreover, the role of technology varies: in some interventions, the technology itself is the vehicle for delivering compassion (e.g., empathic conversational agents), while in others it augments human-delivered care (e.g., blended models). This distinction has implications for how we conceptualise "compassionate technology" in mental health.

5.2 Strengths of the evidence base

The evidence base benefits from a growing number of randomized controlled trials and feasibility studies across diverse populations, including young people, veterans, caregivers, and healthcare professionals. The use of validated instruments such as the Self-Compassion Scale and the Interpersonal Reactivity Index lends credibility to outcome measurement. Several studies have demonstrated long-term follow-ups (e.g., 9 months in Sommers-Spijkerman et al. [2018]), indicating that digital compassion interventions can produce durable effects. The interdisciplinary nature of the field, spanning clinical psychology, human–computer interaction, and artificial intelligence, has enriched both intervention design and evaluation methodologies. Additionally, the synthesis across six

themes reveals that compassionate technology is not confined to a single modality but is explored across apps, virtual reality, chatbots, online forums, and blended formats, suggesting a breadth of innovation.

5.3 Gaps and limitations in the literature

Despite these strengths, several gaps and limitations persist across the literature. High attrition rates remain a pervasive challenge, particularly in app-based and self-guided interventions (e.g., Mak et al. [2018], Aboujaoude et al. [2020]), raising questions about engagement and the representativeness of completers. Many studies lack long-term follow-up beyond three months, leaving the sustainability of effects uncertain. The mechanisms through which digital interventions cultivate compassion are often unexplored, and few studies explicitly test mediators such as fear of compassion or empathic concern. For certain themes, such as blended and hybrid care models, the evidence is notably sparse (only four articles), limiting conclusions about their effectiveness compared to purely digital or face-to-face alternatives. Furthermore, the evidence-card completeness and extraction audit reveals that many articles rely on small samples and uncontrolled designs, and a substantial proportion (11 out of 86 included) were marked as uncertain in the screening process, indicating that the quality of evidence is variable. Another limitation is the inconsistent operationalisation of compassion and empathy across studies; some equate compassion with self-compassion, others with empathy or empathic communication, which complicates cross-study synthesis.

5.4 Implications for future research

Future research should prioritise well-powered, longitudinal randomised controlled trials that incorporate active control conditions and objective behavioural measures of compassion (e.g., prosocial behaviour as in Herrera et al. [2018]). There is a pressing need to investigate mediators and moderators of intervention effects, such as baseline compassion levels, user engagement patterns, and fears of compassion, to identify for whom and under what conditions digital compassionate technologies are most beneficial. Personalization of AI-driven conversational agents (e.g., Abd-Alrazaq et al. [2021]) and the integration of gamification (e.g., Mak et al. [2018]) may help address attrition. For blended care models, rigorous comparative effectiveness trials are needed to clarify the additive value of digital components over face-to-face therapy alone. Additionally, future work should develop theory-driven digital tools specifically for underserved groups such as caregivers, where current interventions focus more on skill-building than on mental health (e.g., Petrovic and Gaggioli [2020]). Finally, adopting common outcome frameworks (e.g., the five elements of compassion from Strauss et al.) would facilitate meta-analytic synthesis and evidence aggregation.

5.5 Limitations of the automated review process

This review employed an automated screening and synthesis pipeline, which introduces several limitations. From the initial 5,876 records downloaded, 3,034 remained after DOI metadata cleaning, and the top 1,000 ranked records were selected for full screening; this ranking process may have excluded relevant studies not captured by the semantic ranking algorithm. Only 200 article cards were generated, of which 86 were included in the final synthesis, with 103 excluded and 11 marked as uncertain. The reliance on automated screening may have introduced systematic biases, for example, over-representing studies with titles or abstracts closely matching the review's search terms. Furthermore, we conducted an evidence-card completeness and extraction audit rather than a formal quality assessment, meaning that risk of bias was not systematically evaluated. The use of a single large language model (Qwen2.5-3B-Instruct) for theme discovery and synthesis may

have limited the range of themes identified, although studies could belong to multiple themes, and the six themes were iteratively refined. Finally, because only 14 of the 86 included articles were based solely on abstracts and metadata, the depth of evidence extraction was constrained for a minority of studies. These limitations should be considered when interpreting the findings, and future reviews may benefit from combining automated approaches with traditional manual screening and quality appraisal.

6 Conclusion

6.1 Overall conclusion

This systematic scoping review mapped the literature on compassionate technology in digital and blended mental health interventions, identifying 6 themes: Self-Compassion Digital Interventions, Empathy-Enabling Technologies, Peer Support and Social Connectedness, Blended and Hybrid Care Models, AI and Conversational Agents for Compassion, and Compassion for Caregivers and Providers. Studies may belong to multiple themes. The review reveals that digital technologies across modalities—from apps and virtual reality to chatbots and online peer platforms—can effectively evoke, enhance, or support compassion and empathy, with positive outcomes for mental well-being and prosocial behavior. However, the field is characterized by heterogeneity in study designs, outcome measures, and intervention fidelity, and significant gaps remain in understanding mechanisms, long-term effects, and real-world implementation.

6.2 Most important evidence patterns

Synthesizing across themes, several cross-cutting patterns emerge. First, there is a consistent contrast between self-report and behavioral or objective outcomes: immersive interventions such as virtual reality perspective-taking produce stronger behavioral effects than self-report differences, suggesting a disconnect between subjective and behavioral empathy Herrera et al. [2018]. Second, engagement and adherence are critical barriers across all digital formats, from app-based self-compassion programs to AI conversational agents and peer platforms Aboujaoude et al. [2020], Garrido et al. [2019]. Third, evaluation methods are shifting toward mixed-methods designs that combine quantitative trials with qualitative and behavioral measures to capture the multifaceted nature of compassion Alvarez-Jimenez et al. [2020], Hargreaves et al. [2018], Baghaei et al. [2019]. Fourth, effectiveness appears enhanced when digital components are combined with human support—as in blended care or guided self-help—indicating that compassionate technology is most impactful when it augments rather than replaces human interaction Boots et al. [2018], Rooney [2020]. Fifth, a notable gap persists in theory-driven intervention design, with many studies lacking explicit grounding in compassion frameworks such as the five elements of compassion.

6.3 Practical and research implications

For practitioners, these findings suggest that digital tools can serve as viable adjuncts or components of compassionate mental health care, but careful attention must be paid to user engagement, personalization, and integration with clinical workflows. Researchers should adopt more rigorous and standardized evaluation designs, including randomized controlled trials with longitudinal follow-up and measures that capture both subjective and behavioral dimensions of empathy and self-compassion. The elements of compassion framework provides a useful conceptual basis for designing

and assessing interventions across the six identified themes. Furthermore, studies examining cost-effectiveness, scalability, and real-world implementation in statutory mental health services are urgently needed to move the field beyond pilot demonstrations.

6.4 Future work

Future research should address the limitations identified in this review. Priority areas include: (a) longitudinal studies to assess sustained effects of digital compassion interventions beyond 6–9 months; (b) investigation of mechanisms of change, such as the role of fears of compassion and user expectations Sommers-Spijkerman et al. [2018]; (c) development and testing of personalization algorithms and adaptive interventions to enhance engagement and reduce attrition Camenga and Kong [2020], Venkatesan et al. [2020]; (d) larger-scale trials for AI conversational agents and blended care models to establish effectiveness and clinical translation Inkster et al. [2018], D'Alfonso et al. [2017]; (e) inclusion of underserved populations and diverse cultural contexts to ensure generalizability; and (f) exploration of how digital compassion technologies can be integrated into statutory mental health services, moving from pilot studies to widespread adoption Rooney [2020]. Collectively, these directions will strengthen the evidence base for compassionate technology and support its responsible implementation in digital and blended mental health care.

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